Reflections

Exam code: 9MA0

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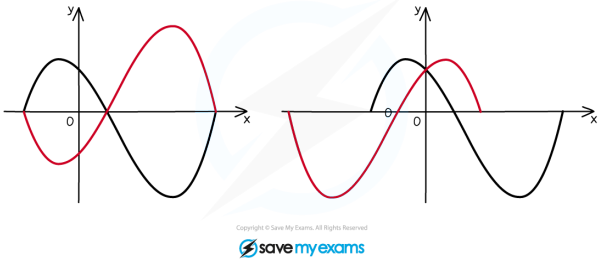
Reflections

What are graph reflections?

- When you alter a function in certain ways, the effects on the graph of the function can be described by geometrical transformations
- With a **reflection** all the points on the graph are reflected in either the ***x*** or the ***y* axis**

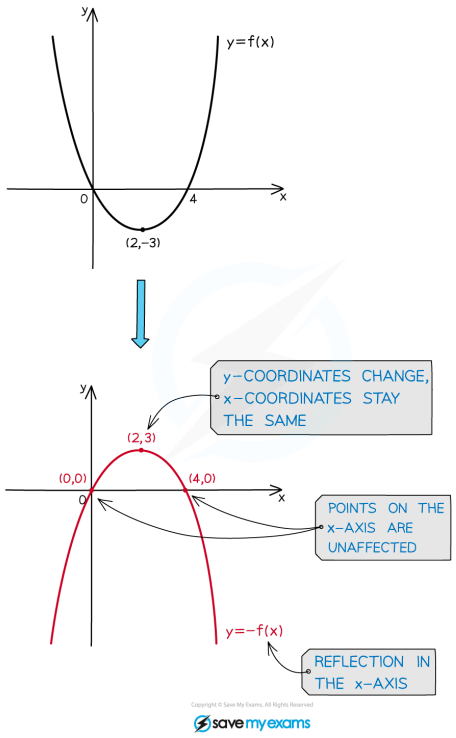


- Any asymptotes of **f(*x*)** are also affected by the reflection (reflect them as you would reflect the function of a straight line)
- If an asymptote is one of the coordinate axes, or is perpendicular to the coordinate axis in which the graph is reflected, it will not be affected

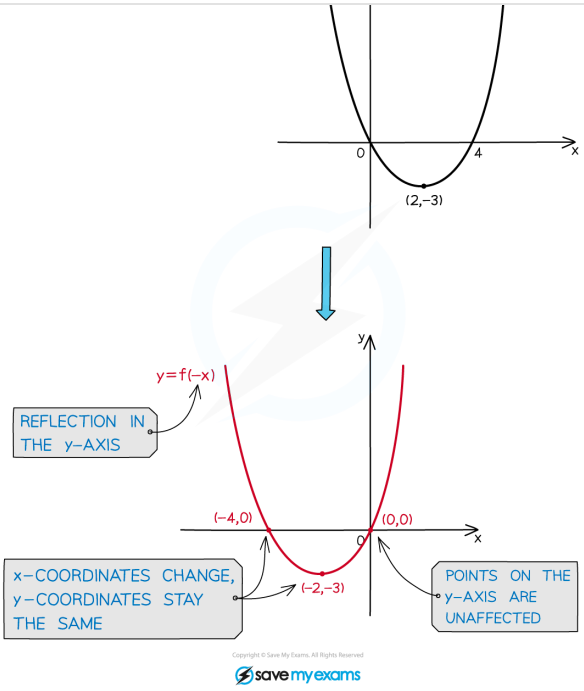
Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

Help me understand

- The graph of $y = -f(x)$ is a reflection in the x axis
  - The x coordinates of points stay the same; y coordinates have their signs flipped (positive to negative, negative to positive)
  - Points **on the x axis** stay where they are
  - All other points are reflected to the other side of the x axis



- The graph of $y = f(-x)$ is a reflection in the y axis
  - The y coordinates of points stay the same; x coordinates have their signs flipped (positive to negative, negative to positive)
  - Points **on the y axis** stay where they are
  - All other points are reflected to the other side of the y axis



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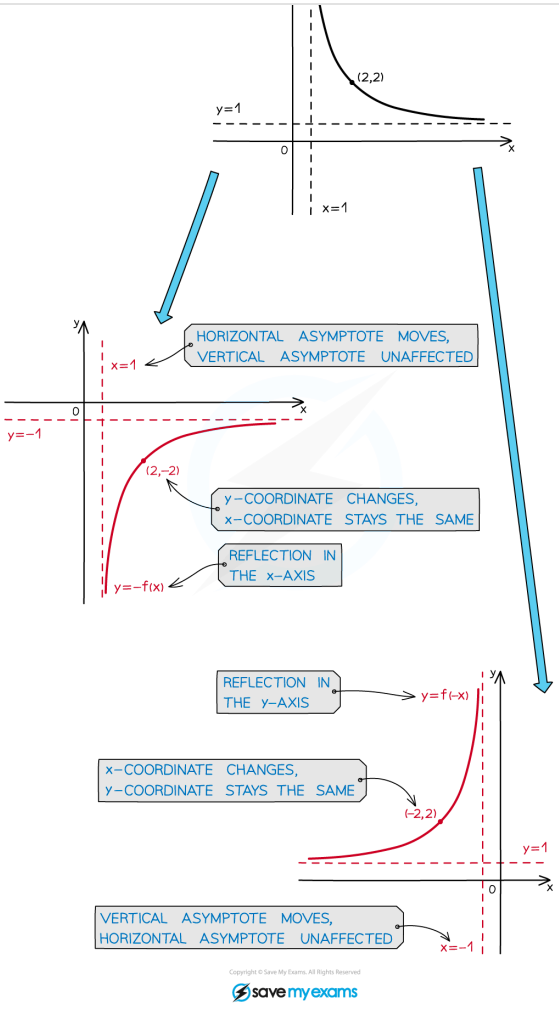

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